CV

CostVision

Manufacturing Should-Cost Intelligence Platform

SHOULD-COST ANALYSIS REPORT

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Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

Motorcycle Triple Clamp / Steering Yoke

Commodity: CASTING · Currency: CNY · FX Rate: 9.0498 to GBP · Operations: 2 · Region: CN

Total Should-Cost	Material	Process	Labour	Margin
¥281.52	**15.3%**	**46.3%**	**12.8%**	**7.4%**

Model Confidence: **Low** · 2 traced operations · 6 data points auditable

Engine Warnings (1)

rawMaterial.materialId: Material rate confidence: Medium

Geometry Provenance — MEASURED (OCCT B-rep kernel)

Volume, weight and every feature were measured from the CAD solid by the Open CASCADE kernel. Measured volume 429.4 cm³; mass for the selected material family 1.151 kg.
Material cost and geometry-derived machining are grounded in the actual solid — not an estimate.

Key Assumptions

Annual volume: 200,000 · Region: CN · Commodity: casting
Alloy / material: LM25 / A356 · Net weight: 1.195 kg (measured)
Material utilisation: 70% · Overhead: 12% of factory base · Margin: 8% of subtotal · Operations: 2

§1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
1. Raw Material	¥43.08	15.3%	
2. Process (Machine)	¥130.25	46.3%	
3. Direct Labour	¥36.12	12.8%	
4. Tooling (amortised)	¥21.27	7.6%	
5. Packaging	¥0.90	0.3%	

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
6. Logistics	¥1.36	0.5%	
Factory Cost	**¥232.98**	**82.8%**	
7. Overhead (SG&A)	¥27.69	9.8%	
Subtotal	**¥260.67**	**92.6%**	
8. Supplier Margin	¥20.85	7.4%	
TOTAL SHOULD-COST	**¥281.52**	**100.0%**	

§2 — Commercial Parameters

Overhead Rate	**12.0%**	Supplier Margin	**8.0%**
Packaging / part	**¥0.90**	Logistics / part	**¥1.36**
Total Tooling Cost	**¥4253393.67**	Amortisation Volume	**200,000 parts**

§3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	**mat-lm25**	ID	
Grade / Specification	**LM25 / A356**		UK Al alloy ingot, Jun 2026. Index-anchored 2026-07 (see CASTING PRICING BASIS). | Regional adj. ×0.970 (exchangeMetal)
Region	**China**		
Net Finished Weight	**1.1948 kg**	kg	Weight in finished part
Gross Weight (stock)	**1.7069 kg**	kg	Net ÷ utilisation ratio
Scrap / Runner Weight	**0.5121 kg**	kg	Gross - Net
Material Utilisation	**70.0%**	%	Benchmark: casting 65–85 % (runners + risers)
Material Price	**¥26.69**	CNY /kg	UK Al alloy ingot, Jun 2026. Index-anchored 2026-07 (see CASTING PRICING BASIS). | Regional adj. ×0.970 (exchangeMetal)
Scrap Recovery Price	**¥4.83**	CNY /kg	
Gross Material Cost	**¥45.55**	CNY	Gross × price/kg
Scrap Credit	**-¥2.47**	CNY	Scrap × recovery price
NET RAW MATERIAL COST	**¥43.08**	CNY	Gross - scrap credit + consumables
Data Confidence	**Medium**		
Effective Date	**2026-07**		

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
Gravity Die Casting	Gravity Die Casting Machine	¥178.02	4.95	1	80.0%	**¥18.35**
CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face	CNC VMC 3-axis	¥254.30	22.44	1	85.0%	**¥111.90**

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured]						
TOTAL						**¥130.25**

4B Labour Detail

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
Gravity Die Casting	Skilled Machinist	¥71.49	1	4.95	92.0 %	¥6.41	**¥24.76**
CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured]	Skilled Machinist	¥71.49	1	22.44	90.0 %	¥29.71	**¥141.61**
TOTAL						**¥36.12**	**¥166.37**

4C Machined Features — Geometry Audit

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Hole / bore	Ø8.0 × 18 blind	2	Drilling (blind)	1.20	¥5.10	Yes
Hole / bore	Ø8.0 × 23 blind	2	Drilling (blind)	1.44	¥6.09	Yes
Hole / bore	Ø10.5 × 13 blind	2	Drilling (blind)	1.03	¥4.36	Yes
Hole / bore	Ø14.0 × 10 blind	2	Drill + bore (blind)	1.80	¥7.63	Yes
Hole / bore	Ø22.0 × 40 blind	2	Drill + bore (blind)	4.17	¥17.67	Yes
Hole / bore	Ø24.0 × 16 blind	1	Drill + bore (blind)	1.14	¥4.83	Yes
Hole / bore	Ø43.0 × 8 blind	1	Drill + bore (blind)	0.37	¥1.58	Yes
Hole / bore	Ø50.0 × 13 blind	1	Drill + bore (blind)	1.25	¥5.28	Yes
Hole / bore	Ø50.0 × 32 blind	2	Drill + bore (blind)	4.40	¥18.65	Yes
Face	2915 mm²	2	Face milling (facing)	1.13	¥4.79	Yes
Face	2551 mm²	2	Face milling (facing)	1.04	¥4.40	Yes
Face	1986 mm²	2	Face milling (facing)	0.90	¥3.80	Yes
Face	1423 mm²	2	Face milling (facing)	0.76	¥3.20	Yes
Face	1357 mm²	1	Face milling (facing)	0.37	¥1.57	Yes
Face	868 mm²	1	Face milling (facing)	0.31	¥1.31	Yes

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Face	698 mm²	2	Face milling (facing)	0.57	¥2.44	Yes
Face	695 mm²	2	Face milling (facing)	0.57	¥2.43	Yes
Pocket	2915 mm² × 22	2	Pocket milling (rough + finish)	22.97	—	No
Pocket	1986 mm² × 26	2	Pocket milling (rough + finish)	18.72	—	No
TOTAL costed		**29**		**22.44**	**¥95.11**	**Yes**

4 detected feature(s) NOT costed

4 machinable feature instance(s) (41.7 min) were detected in the geometry but are NOT included in this cost — on a near-net part they may be as-cast. If they are machined, tick them in the Machined-Features panel and re-cost; this could add up to ¥176.71.

§5 — Machine Rate Buildup

CNC VMC 3-axis [mach-vmc3] · Confidence: Low · UK Tier-2 3-axis VMC (HAAS VF/VM class). Target £55/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥273755.66	¥85.55	4,000 hr/yr available
Maintenance	¥149321.27	¥46.66	
Energy	¥77119.81	¥24.10	
Floor Space	¥59728.51	¥18.67	
Indirect Support	¥149321.27	¥46.66	
Finance Cost	¥104524.89	¥32.66	
TOTAL — CNC VMC 3-axis	¥813771.39	¥254.30	Effective hours: 3200/yr

Gravity Die Casting Machine [grav-die-cast-std] · Confidence: Low · Standard gravity die casting machine, UK foundry benchmark | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @50% util	Notes
Depreciation	¥139366.52	¥69.68	4,000 hr/yr available
Maintenance	¥69683.26	¥34.84	
Energy	¥27542.79	¥13.77	
Floor Space	¥39819.00	¥19.91	
Indirect Support	¥44796.38	¥22.40	
Finance Cost	¥34841.63	¥17.42	
TOTAL — Gravity Die Casting Machine	¥356049.58	¥178.02	Effective hours: 2000/yr

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.

Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

§6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf.
material.pricePerKg	2.9488	£/kg	UK Al alloy ingot, Jun 2026. Index-anchored 2026-07 (see CASTING PRICING BASIS). |	mat-lm25	Medium

Field	Value	Unit	Source / Reference	Rate ID	Conf.
			Regional adj. ×0.970 (exchangeMetal)		
material.scrapRecoveryPricePerKg	0.5335	£/kg	UK Al alloy ingot, Jun 2026. Index-anchored 2026-07 (see CASTING PRICING BASIS). | Regional adj. ×0.970 (exchangeMetal)	mat-lm25	Medium
Gravity Die Casting.machineRatePerHr	19.6717	£/hr	Standard gravity die casting machine, UK foundry benchmark | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	grav-die-cast-std	Low
Gravity Die Casting.labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured].machineRatePerHr	28.1005	£/hr	UK Tier-2 3-axis VMC (HAAS VF/VM class). Target £55/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-vmc3	Low
CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured].labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low

§7 — Should-Cost with Uncertainty

Monte-Carlo · Low confidence

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
¥281.52 ± 17.8%	¥232.94	¥278.37	¥333.30	wide	14%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 4 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread; some detected features not costed (see §4C).

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.
Import duty and international freight (regional table is Ex-Works).
Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).
Heat-treat (T6/T7), HIP and 100% NDT/X-ray are NOT in this cost — add them for structural / safety-critical parts.
Trim/degate, shot-blast/deflash and impregnation are only included where shown as operations.

§8 — Sensitivity Analysis

± 10% driver swing · ranked by ¥ impact

Cost Driver	Base	-10% cost	+10% cost	Range ¥
CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured]: Cycle Time	0.37hr	-6.1%	+6.1%	**¥34.26**
CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured]: Machine Rate	¥254.30/hr	-4.8%	+4.8%	**¥27.07**
Material: LM25 / A356	¥26.69/kg	-2%	+2%	**¥11.02**
Material utilisation	0.7	+1.8%	-1.5%	**¥9.12**
Gravity Die Casting: Labour Rate	¥71.49/hr	-1.6%	+1.6%	**¥8.74**
CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured]: Labour Rate	¥71.49/hr	-1.6%	+1.6%	**¥8.74**
Gravity Die Casting: Cycle Time	0.08hr	-1.1%	+1.1%	**¥5.99**
Overhead %	12%	-1.1%	+1.1%	**¥5.98**

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	¥48.95	¥236.82	¥130.03	¥38.67	¥36.91	**¥491.39**	¥0.94	**¥514.47**	+83%
Germany (EUR)	¥50.42	¥248.66	¥210.15	¥40.60	¥49.13	**¥598.96**	¥1.08	**¥626.80**	+123%
France (EUR)	¥49.93	¥217.88	¥151.05	¥35.57	¥38.76	**¥493.18**	¥1.08	**¥516.56**	+83%
Spain (EUR)	¥48.95	¥208.40	¥95.22	¥34.03	¥29.83	**¥416.44**	¥1.08	**¥436.53**	+55%
Poland (PLN)	¥47.48	¥170.51	¥59.11	¥27.84	¥21.05	**¥325.99**	¥1.12	**¥342.15**	+22%
Turkey (TRY)	¥44.06	¥142.09	¥32.84	¥23.20	¥15.34	**¥257.53**	¥1.22	**¥270.76**	-4%
China (CNY)	¥43.08	¥130.25	¥36.12	¥21.27	¥27.69	**¥258.40**	¥1.36	**¥281.52**	Base
India (INR)	¥44.06	¥123.15	¥22.99	¥20.11	¥12.30	**¥222.60**	¥1.40	**¥234.34**	-17%
Mexico (MXN)	¥46.51	¥142.09	¥38.09	¥23.20	¥15.83	**¥265.72**	¥1.26	**¥279.27**	-1%
United States (USD)	¥48.95	¥201.30	¥170.75	¥32.87	¥35.02	**¥488.89**	¥1.17	**¥512.04**	+82%

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

15.35 kgCO2e · 12.85 /kg · cradle-to-gate

Material	**10.94 kgCO2e**	Material factor	**8.60 kg/kg (Aluminium (primary mix))**
Process	**4.22 kgCO2e**	Grid intensity	**0.550 kg/kWh · 7.68 kWh**
Logistics	**0.19 kgCO2e**	Total	**15.35 kgCO2e**

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

§11 — Cost Intelligence Insights

4 findings · ~17% combined saving

[WARNING] Machining / process cost above benchmark

Finding	Process cost at 46.3% of total is above the benchmark range of 8–16%. Cycle time or machine rate is elevated.
Impact	High · up to 15% potential saving
Benchmark	Process %: yours 46.3% vs industry 8–16%
Action 1	Reduce cycle time via feeds/speeds optimisation and toolpath strategy review
Action 2	Combine multiple operations into one setup to eliminate re-fixture time

[WARNING] Elevated process cost — possible high reject/scrap rate

Finding	Process cost is 46.3% vs benchmark max of 16%. For castings, this may indicate a high reject rate (>5%) inflating effective machine time, poor OEE on the casting cell, or cycle time above industry benchmark for this alloy/weight range.
Impact	Medium · up to 8% potential saving
Action 1	Audit actual first-pass yield (FPY) data from the foundry — target >96% for HPDC Al
Action 2	Root-cause the dominant reject type: porosity, dimensional, surface, or cold shut

[INFO] Material cost is below benchmark range

Finding	Material at 15.3% is below the typical 40–65% range. Conversion cost is the dominant driver.
Impact	Low
Benchmark	Material %: yours 15.3% vs industry 40–65%
Action 1	Focus optimisation efforts on process and labour efficiency

[INFO] Confirm post-casting scope — no finishing operations in this cost

Finding	The operation list carries no heat treatment, shot blasting or fettling line. If the casting form's service adders were set they are already in the material bucket; if not, structural castings typically add 8–18% for T5/T6 heat treatment (£1.20–2.80/kg), shot blast (£0.15–0.40/part), impregnation (£0.80–1.80/part) and deburring (£0.10–0.60/part).
Impact	Medium
Action 1	Check the derivation trace: heat-treat/descale/NDT adders appear there when they were costed
Action 2	If genuinely absent, set the casting service adders (heat treat, shot blast, impregnation) and re-cost

§12 — Cost-Reduction Opportunities (ranked by saving)

6 opportunities · combined ¥45.89/part (~16%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/part	%	Timeframe	Risk
1	Process & Automation	**Multi-Cavity / Multi-Up Tooling**	Every cycle currently makes one part. A 2-cavity (or family) tool halves the machine and labour minutes per part against roughly 60–80% more tooling NRE.	¥33.78	12.0 %	Medium Term	Medium
2	Process & Automation	**Attack the Bottleneck: "CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured]"**	One operation — CNC Machining — 29 features (2×Ø8.0×18, 2×Ø8.0×23, 2×Ø10.5×13, 2×Ø14.0×10, 2×Ø22.0×40, 1×Ø24.0×16, 1×Ø43.0×8, 1×Ø50.0×13, 2×Ø50.0×32, 2× face 2915mm², 2× face 2551mm², 2× face 1986mm², 2× face 1423mm², 1× face 1357mm², 1× face 868mm², 2× face 698mm², 2× face 695mm²) [geometry-measured] — carries 85% of the conversion cost. Improving anything else first is working on the wrong problem.	¥22.52	8.0%	Medium Term	Medium
3	Material	**Improve Material Utilisation via Near-Net-Shape**	Current utilisation at 70%. Near-net-shape process or improved nesting targets 80–90%.	¥21.11	7.5%	Medium Term	Medium
4	Design & Geometry	**Design Review for DFM Compliance**	Formal DFM review with manufacturing engineering to identify part features that increase cost without functional benefit.	¥14.08	5.0%	Quick Win	Low
5	Commercial & Sourcing	**Annual Volume Re-Commitment for Better Pricing**	Provide 12-month rolling volume forecast to supplier to secure better unit pricing and tooling amortisation.	¥8.45	3.0%	Quick Win	Low
6	Sustainability & Energy	**Energy Productivity Programme**	This process is energy-intensive (melt, cure, oven or heat-treat content). Energy sits inside the machine rate and overhead — and it is the fastest-moving input cost of the decade.	¥8.45	3.0%	Medium Term	Low

§13 — Opportunity by Category

5 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Process & Automation	2	Multi-Cavity / Multi-Up Tooling	¥33.78	¥56.30
Material	1	Improve Material Utilisation via Near-Net-Shape	¥21.11	¥21.11
Design & Geometry	1	Design Review for DFM Compliance	¥14.08	¥14.08
Commercial & Sourcing	1	Annual Volume Re-Commitment for Better Pricing	¥8.45	¥8.45
Sustainability & Energy	1	Energy Productivity Programme	¥8.45	¥8.45

§14 — Inputs to Confirm

No saving is claimed against these

Item	What the costing assumed	How to close it
Unusually low material content — verify alloy grade	Material at 15.3% of part cost is below the 35% floor for castings. Verify alloy grade and net weight inputs.	Validate material weight and alloy price inputs against purchase orders

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Design Review for DFM Compliance
- Annual Volume Re-Commitment for Better Pricing